

Test Name : Monaka HPC AI

Composition

Sections 5	Questions 26	Time 45 min	
Software Engineering Basics	5	6 min	
Data structures	5	6 min	
Python and C++	5	7 min	
Machine Learning/Deep Learning	10	11 min	
Backend Coding	1	15 min	

Section: 1 of 5 | Software Engi...



Question: 1 of 5



Choose the correct option

Coupling and cohesion can be represented using a _____ .

- ☐ cause-effect graph
- ☐ dependence matrix
- ☐ Structure chart
- ☐ SRS

Clear

Next

Section: 1 of 5 | Software Engi...



Question: 2 of 5



What does CI in CI/CD stand for?

Choose the correct option

- ☐ Continuous Input
- ☐ Continuous Integration
- ☐ Continuous Improvement
- ☐ Continuous Installation

Clear

Next

Question: 3 of 5



What is the purpose of A/B testing in MLOps?

Choose the correct option

- ☐ A process for comparing the performance of multiple machine learning models
- ☐ A process for evaluating the impact of changes to machine learning models in production
- ☐ A process for evaluating the impact of changes to data used in machine learning models
- ☐ None of the options

Section: 1 of 5 | Software Engi...



Question: 4 of 5



What does IaC stand for in DevOps?

Choose the correct option

- ☐ Infrastructure as Computer
- ☐ Integration and Configuration
- ☐ Infrastructure as Code
- ☐ Input and Compute

Clear

Next

Question: 5 of 5



In DevOps, what does "shift left" refer to?

Choose the correct option

- ☐ Delaying testing to the end of the development cycle
- ☐ Left-aligning code
- ☐ Outsourcing operations
- ☐ Incorporating operations earlier in the development cycle

[Clear](#) This is the last question, submit this section & go to next section.

Question: 1 of 5

Current question



Choose the correct option

What will be the output of the following C++ program?

```
#include <iostream>
#include <string>
using namespace std;
int main(int argc, char const *argv[])
{
    char s1[6] = "Hello";
    char s2[6] = "World";
    char s3[12] = s1 + " " + s2;
    cout<<s3;
    return 0;
}
```

- ☐ Hello
- ☐ World
- ☒ Error
- ☐ Hello World

[Clear](#)[Next](#)

Question: 2 of 5



Choose the correct option

A is a matrix of size $n \times n$. Consider the following program. What is the expected output?

```
void fun(int A[][N])
{
    for (int i = 0; i < N; i++)
        for (int j = i + 1; j < N; j++)
            swap(A[i][j], A[j][i]);
}
```

- ☒ Matrix A
- ☐ Diagonal Of matrix A
- ☐ Transpose of matrix A
- ☐ None

[Clear](#)[Next](#)

Question: 3 of 5



Choose the correct option

What is the output of the following code?

```
int a = 1;
int b = 2;
cout << (a++ + b);
```

3

4

Syntax error

2

Clear

Next

Question: 4 of 5



Predict the output

```
y = 8
```

```
z = lambda x : x * y
```

```
print (z(6))
```

Choose the correct option

48

14

64

None of the options

Clear

Next

Question: 5 of 5



What is the output of the following C++ program?

```
#include<iostream>
using namespace std;
```

```
class Base
{
public:
    virtual void show() { cout<<" In Base \\n"; }
};
```

```
class Derived: public Base
{
public:
    void show() { cout<<"In Derived \\n"; }
};
```

```
int main(void)
{
    Base *bp = new Derived;
    bp->Base::show();
    return 0;
}
```

Choose the correct option

- ☐ In Base
- ☐ In Derived
- ☐ Compiler Error
- ☐ Runtime Error

[Clear](#) This is the last question, submit this section & go to next section.

Question: 1 of 10



Choose the correct option

What is the function of an Artificial Intelligence "Agent"?

- ☐ Mapping of goal sequence to an action
- ☐ Work without the direct interference of the people
- ☐ Mapping of precept sequence to an action
- ☐ Mapping of environment sequence to an action

Clear

Next

Question: 2 of 10



Choose the correct option

What is/are the limitations of deep learning?

- A) Data labeling
- B) Obtain huge training datasets

- A
- B
- A and B
- None of the options

Clear

Next

Question: 3 of 10



Choose the correct option

Identify the activation function output which is zero centered.

- Hyperbolic tangent
- Rectified linear unit
- Sigmoid
- Softmax

Clear

Next

Question: 4 of 10



Choose the correct option

What is the purpose of skip connections in deep convolution networks like ResNet?

- To reduce the number of layers in the network
- To skip training process for certain layers
- To combat the vanishing gradient problem and facilitate training of very deep neural networks
- To reduce computational complexity

Clear

Next

Question: 5 of 10



Choose the correct option

What are the general limitations of the backpropagation rule?

- Slow convergence
- Local minimum problem
- Scaling
- All of the options

Clear

Next

Question: 6 of 10



Choose the correct option

Which evaluation metric is commonly used for binary classification problems to measure the trade-off between precision and recall?

- Accuracy
- F1-score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Clear

Next

Question: 7 of 10



Choose the correct option

Which machine learning algorithms are based on the idea of bagging?

- Decision Tree
- Random Forest
- Regression
- Classification

Clear

Next

Question: 8 of 10



Choose the correct option

Which function can be used as an activation function in the output layer if you wish to predict the probabilities of n classes ($p_1, p_2, \dots, p_k$) such that the sum of p over all n equals 1?

- Softmax
- Relu
- Sigmoid
- Tanh

Clear

Next



Question: 9 of 10



Choose the correct option

_____ is a method in statistics of inference used to update the probability estimate for a hypothesis as additional evidence is learned.

- Bayesian Inference
- Inductive theory
- Markov Theory
- Information Field theory

Clear

Next



Question: 10 of 10



Choose the correct option

Which option is not a supervised learning algorithm?

- PCA
- Linear Regression
- Naïve Bayes
- Decision Tree

[Clear](#) This is the last question, submit this section & go to next section.